higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

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AUGUST EXAMINATION

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

2 August 2016 (X-Paper)
9:00–12:00

Scientific calculators may be used.

This question paper consists of 6 pages and 1 formula sheet of 5 pages.

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
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MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL intermediate steps and simplify where possible.
5. ALL final answers must be rounded off to THREE decimals.
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Questions must be answered in blue or black ink.
8. Write neatly and legibly.

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QUESTION 1

1.1 Determine $\ln y$ if:

1.1.1 $\ln y = \lim\limits_{x \to -0} \dfrac{x}{\ln \sin 3x}$ (3)

1.1.2 Now, determine the value of y (1)

1.2 Determine $\lim\limits_{x \to \infty} x^2.e^{-x}$ (3)

[7]

QUESTION 2

2.1 Given: $f(x) = \dfrac{2-x}{3-2x}$

Determine the derivative $f'(x)$ using first principles. (4)

2.2 Determine $\dfrac{dy}{dx}$ in each of the following cases (simplification is NOT required):

2.2.1 $y = \dfrac{1-x}{\sqrt{\tan x}}$ (4)

2.2.2 $y = [\ln(\cos ec 2x)]^5$ (3)

2.2.3 $y = 2e^{\sin 5x}.\sqrt[3]{x}$ (2)

2.3 Determine $\dfrac{dy}{dx}$ with the aid of logarithmic differentiation if $y = (9-x^3)^{\ln x}$ (4)

2.4 Given: $y = \arcsin x$

2.4.1 Derive a formula to determine $\dfrac{dy}{dx}$ if $y = \arcsin x$ (2)

2.4.2 Hence, determine $\dfrac{dy}{dx}$ if $y = \arcsin(5.10^x)$ (2)

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2.5 Given: the implicit function $x^2 + 2 = y^3.\cot 3x$

Calculate the derivative $\frac{dy}{dx}$ (4)

[25]

QUESTION 3

3.1 Given: $f(x) = x(x^2 - 3x - 9)$

3.1.1 Determine the coordinates of the point of inflection of $f(x)$ (2)

3.1.2 Draw up a table of values for x and $f(x)$ with x ranging from $x = -1$ to $x = 3$ (2)

3.1.3 Draw a neat graph of $f(x)$ between the values in QUESTION 3.1.2 if the turning points are given as (3;–27) and (–1;5) (2)

3.1.4 An estimated root of the graph is –0,1

Use Taylor's/Newton's method to determine a better approximation of this root, showing that the root tends to zero. (3)

3.2 Let V be the volume of a cylinder having height h and radius r.

Assume that h and r vary with time.

3.2.1 How are $\frac{dV}{dt}$, $\frac{dr}{dt}$ and $\frac{dh}{dt}$ related? (2)

3.2.2 At a certain instant the height is 30 cm and it decreases at 1 cm/s while the radius of 11 cm increases at 0,5 cm/s.

How fast is the volume changing at that instant?

HINT: $V = \pi r^2 h$ (5)

3.3 The sum of the diameter and height of a cylinder is 6 cm.

Calculate the dimensions of the cylinder to ensure a maximum volume. (5)

[21]

QUESTION 4

4.1 Given: $\int \frac{4x}{100 - x^2} = \frac{A}{(10 - x)} + \frac{B}{(10 + x)} dx$

4.1.1 Determine the numerical values of A and B. (2)

4.1.2 Hence, calculate $\int_1^3 \frac{4x}{100 - x^2} dx$ (3)

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4.2 Determine $\int y dx$ in each of the following cases:

4.2.1 $y = x^{-2}\sin\frac{1}{x}$

4.2.2 $y = \cos^2 5x$

4.2.3 $y = \cos 12x.\sin 3x$

4.2.4 $y = \frac{dx}{\sqrt{25-25x^2}}$

(4 × 3) (12)

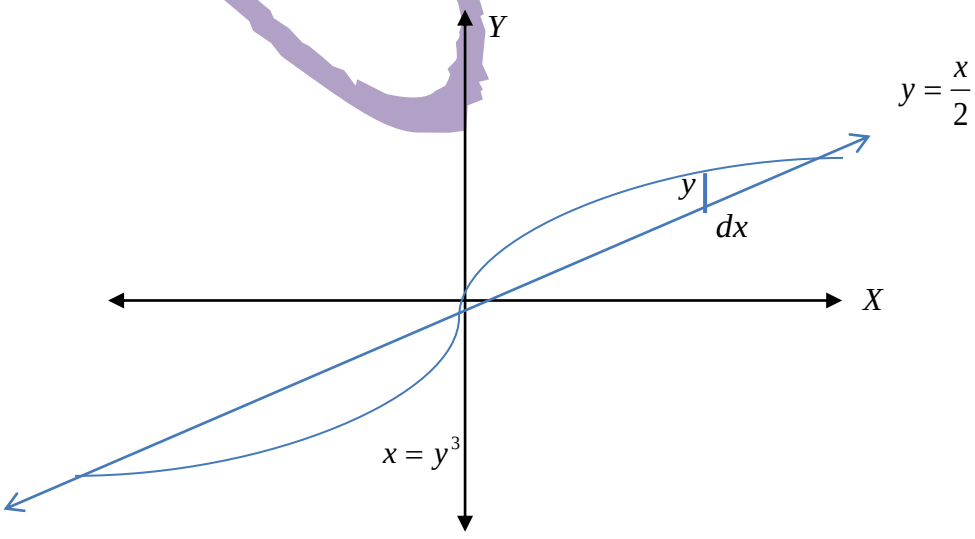
4.3 Given: $\int f(x).g'(x) = f(x).g(x) - \int f'(x).g(x)dx$

Determine: $\int x^{13}\ln 2x \ dx$ (4)

[21]

QUESTION 5

5.1 The sketch shows the area bounded by the curve $x = y^3$ and $y = \frac{x}{2}$



5.1.1 Prove that the points of intersection are:

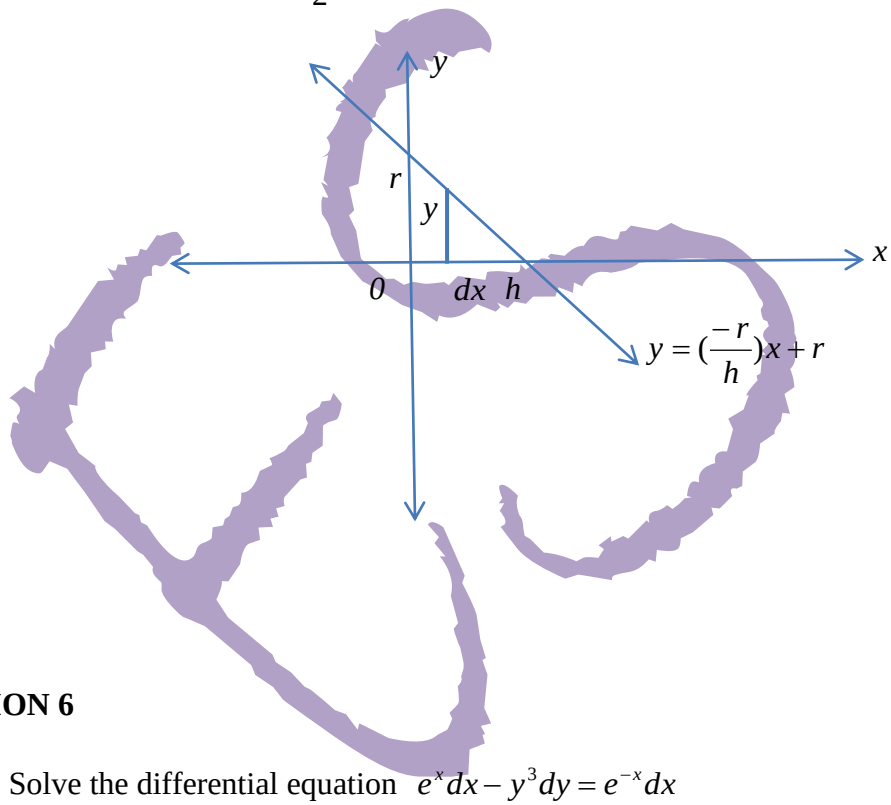
(0;0), (2,828;1,414) and (–2,828;–1,414) (3)

5.1.2 Calculate the area of the bounded region in the first quadrant. (4)

5.1.3 Calculate the volume generated when the area in QUESTION 5.1.1 rotates about the *X*-axis. (4)

5.2 Calculate the value of $\int_1^2 \frac{2x^2}{5x^3}dx$ (3)

5.3 Show that the volume of the figure formed when the line $y = (\frac{-r}{h})x + r$ rotates about the *X*-axis equals $V = \frac{\pi r h}{2}$



(4)
[18]

QUESTION 6

6.1 Solve the differential equation $e^x dx - y^3 dy = e^{-x} dx$ (4)

6.2 Determine the general solution of the following differential equation:

$(1+x)\sin y.dy = \cos y.dx$

(4)
[8]

TOTAL: 100

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MATHEMATICS N5

FORMULA SHEET

Any other applicable formula may also be used.

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \text{cosec}^2 x$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$\tan 2A = \dfrac{2 \tan A}{1 - \tan^2 A}$

$\sin^2 A = \frac{1}{2} - \frac{1}{2} \cos 2A$

$\cos^2 A = \frac{1}{2} + \frac{1}{2} \cos 2A$

$\sin (A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$

$\tan (A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

$\sin A \cos B = \frac{1}{2} [\sin (A + B) + \sin (A - B)]$

$\cos A \sin B = \frac{1}{2} [\sin (A + B) - \sin (A - B)]$

$\cos A \cos B = \frac{1}{2} [\cos (A + B) + \cos (A - B)]$

$\sin A \sin B = \frac{1}{2} [\cos (A - B) - \cos (A + B)]$

$\tan x = \dfrac{\sin x}{\cos x}$; $\sin x = \dfrac{1}{\text{cosec}\, x}$; $\cos x = \dfrac{1}{\sec x}$

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BINOMIAL THEOREM

$$(x+h)^n = x^n + nx^{n-1}\,h + \frac{n(n-1)}{2!}x^{n-2}\,h^2 + \ldots$$

DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x)\,.\,v(x)$$

$$\frac{dy}{dx} = u\,.\,\frac{dv}{dx} + v\,.\,\frac{du}{dx}$$

$$= u \cdot v' + v \cdot u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v\,.\,\frac{du}{dx} - u\,.\,\frac{dv}{dx}}{v^2}$$

$$= \frac{v \cdot u' - u \cdot v'}{v^2}$$

CHAIN RULE

$$y = f(u(x))$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

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$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1} + c$
a	0	$ax + c$
e^x	e^x	$e^x + c$
a^x	$a^x.\ln a$	$\frac{a^x}{\ln a} + c$
$\log_e x$	$\frac{1}{x}$	—
$\log_a x$	$\frac{1}{x \ln a}$	—
$\sin x$	$\cos x$	$-\cos x + c$
$\cos x$	$-\sin x$	$\sin x + c$
$\tan x$	$\sec^2 x$	$\ln(\sec x) + c$
$\cot x$	$-\text{cosec}^2 x$	$\ln(\sin x) + c$
$\sec x$	$\sec x \cdot \tan x$	$\ln[\sec x + \tan x] + c$
$\text{cosec } x$	$-\text{cosec } x \cdot \cot x$	$\ln(\text{cosec } x - \cot x) + c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	—
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	—
$\tan^{-1} x$	$\frac{1}{1+x^2}$	—
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	—
$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$	—
$\text{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$	—

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$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
$\frac{1}{\sqrt{a^2 - x^2}}$	—	$\sin^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{a^2 + x^2}$	—	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{x\sqrt{x^2 - a^2}}$	—	$\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$
$\sqrt{a^2 - x^2}$	—	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{2}\sqrt{a^2 - x^2} + c$
$\frac{1}{x^2 - a^2}$	—	$\frac{1}{2a} ln\left(\frac{x - a}{x + a}\right) + c$
$\frac{1}{a^2 - x^2}$	—	$\frac{1}{2a} ln\left(\frac{a + x}{a - x}\right) + c$

INTEGRATION

$\int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int f'(x)\, g(x)\, dx$

$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c$

$\int [f(x)]^n f'(x)\, dx = \frac{[f(x)]^{n+1}}{n+1} + c$

$\frac{f(x)}{(ax + b)(cx + d)} = \frac{A}{ax + b} + \frac{B}{cx + d}$

$\frac{f(x)}{(x + a)^n} = \frac{A}{(x + a)} + \frac{B}{(x + a)^2} + \frac{C}{(x + a)^3} + \cdots \frac{Z}{(x + a)^n}$

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APPLICATIONS OF INTEGRATION

AREAS

$$A_x = \int_a^b y\,dx\,;\; A_x = \int_a^b (y_1 - y_2)\,dx$$

$$A_y = \int_a^b x\,dy\,;\; A_y = \int_a^b (x_1 - x_2)\,dy$$

VOLUMES

$$V_x = \pi\int_a^b y^2\,dx\,;\; V_x = \pi\int_a^b \left(y_1^2 - y_2^2\right)dx$$

$$V_y = \pi\int_a^b x^2\,dy\,;\; V_y = \pi\int_a^b \left(x_1^2 - x_2^2\right)dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2\,dA\,;\; I_y = \int_a^b r^2\,dA$$

MOMENTS OF INERTIA

Mass = density × volume

$M = \rho V$

DEFINITION: $I = m\,r^2$

GENERAL: $I = \int_a^b r^2\,dm = \rho\int_a^b r^2\,dV$

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